

# Hao Chen

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| CONTACT      | <a href="#">Google Scholar</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">Homepage</a>            | <a href="#">Linkedin</a> | <a href="mailto:chen4433@purdue.edu">chen4433@purdue.edu</a> |
| EDUCATION    | <b>Purdue University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2022 - Present                      |                          |                                                              |
|              | Ph.D. in Chemical Engineering (emphasis on ML/OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Advisor: Dr. <a href="#">Can Li</a> |                          |                                                              |
|              | <b>University of Cambridge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2020 - 2021                         |                          |                                                              |
|              | M.Phil. in Chemical Engineering and Biotechnology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                          |                                                              |
|              | <b>University of Nottingham</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2016 - 2020                         |                          |                                                              |
|              | B.Eng. in Chemical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |                          |                                                              |
| APPOINTMENTS | <b>Amazon</b> , Seattle, WA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026                                |                          |                                                              |
|              | Applied Scientist Intern, Amazon Pharmacy Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Supervisor: Dr. Xin Tang            |                          |                                                              |
|              | <ul style="list-style-type: none"><li>• Built an agent harness that advances optimization explainability, orchestrating filesystem and shell tools, a structured context system, and an asynchronous runtime for large-scale models.</li><li>• Onboarded 3 key models supporting AP’s core services and designed an evaluation pipeline.</li><li>• Benchmarked the harness against a Claude Agent SDK baseline, achieving +36.5 pts success-rate gain at 1.4x lower latency and 56% fewer input tokens.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                          |                                                              |
|              | <b>Amazon</b> , Bellevue, WA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2025                                |                          |                                                              |
|              | Applied Scientist Intern, Last Mile Planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Supervisor: Dr. Jin Ye              |                          |                                                              |
|              | <ul style="list-style-type: none"><li>• Developed an agentic system that interacts with optimization models at runtime and produces trustworthy natural-language explanations for business users without scientist mediation.</li><li>• Delivered the first production integration on the route-allocation model behind delivery service partners handling over 90% of Amazon’s customer packages.</li><li>• Now scaled to 5 models across Last Mile RaP, onboarded by the team and in daily use.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                     |                          |                                                              |
| PUBLICATIONS | <div><div>[1] <b>H Chen</b>, C Qian, C Morris, A Lodi, C Li. Solving Max-Cut to Global Optimality via Feasibility-Preserving Graph Neural Networks. <i>arXiv</i> (2026).</div><div>[2] A Ramanujam, AE Youmi, <b>H Chen</b>, SM Kompalli, AS Ahluwalia, S Pal, DJ Papageorgiou, C Li. SafeOR-Gym: A Benchmark Suite for Safe Reinforcement Learning Algorithms on Practical Operations Research Problems. <i>Transactions on Machine Learning Research</i> (2026).</div><div>[3] H.-H Chiu, A Varghese, K Shao, Y.-C Lu, R Nahar, H Chen, Q Deng, X Bao, C Li. scChat: A large language model-powered co-pilot for contextualized single-cell RNA sequencing analysis. <i>AIChe Journal</i> (2026).</div><div>[4] H Kim, H Chen, C Li, JM Lee. TalkToAgent: A multi-agent LLM Framework for natural language explanation of reinforcement learning policies. <i>Computers &amp; Chemical Engineering</i> (2026).</div><div>[5] GE Constante-Flores, <b>H Chen</b>, C Li. Enforcing Hard Linear Constraints in Deep Learning Models with Decision Rules. <i>Advances in Neural Information Processing Systems</i> 38 (2025).</div><div>[6] <b>H Chen</b>, GE Constante-Flores, KSI Mantri, SM Kompalli, AS Ahluwalia, C Li. OptiChat: Bridging Optimization Models and Practitioners with Large Language Models. <i>INFORMS Journal on Data Science</i> (2025).</div><div>[7] A Khan, R Nahar, H Chen, GE Constante-Flores, C Li. FaultExplainer: Leveraging large language models for interpretable fault detection and diagnosis. <i>Computers &amp; Chemical Engineering</i> (2025).</div><div>[8] <b>H Chen</b>, GE Constante-Flores, C Li. Physics-informed neural networks with hard linear equality constraints. <i>Computers &amp; Chemical Engineering</i> (2024).</div></div> |                                     |                          |                                                              |

- [9] **[Best Paper Award]** H Chen, GE Constante-Flores, C Li. Diagnosing infeasible optimization problems using large language models. *INFOR: Information Systems and Operational Research* (2024).
- [10] Y Yan, WI Shin, H Chen, S.-M Lee, S Manickam, S Hanson, H Zhao, E Lester, T Wu, CH Pang. A recent trend: application of graphene in catalysis. *Carbon Letters* (2021).

#### CONFERENCE PRESENTATIONS

Chen, H. (2026). Solving Max-Cut to Global Optimality via Feasibility-Preserving Graph Neural Networks. Paper presented at the *2026 INFORMS Annual Meeting*, San Francisco, CA

Chen, H. (2026). Neural Branch-and-Bound Solver for Global Optimization of Max Cut. Paper presented at the *2026 AIChE Annual Meeting*, Minneapolis, MN

Chen, H. (2025). Deep Learning Models with Hard Linear Inequality Constraints. Paper presented at the *2025 AIChE Annual Meeting*, Boston, MA **[CAST Director's Student Presentation Award Finalist]**

Chen, H. (2024). Self-supervised Learning for Constrained Optimization with Hard Linear Constraints. Paper presented at the *2024 INFORMS Annual Meeting*, Seattle, WA

Chen, H. (2024). GPU Accelerated Approximation Algorithm for Multi-Parametric Linear Programming. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Chen, H. (2024). Physics-Informed Neural Networks with Hard Linear Equality Constraints. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Chen, H. (2024). Diagnosing Infeasible Optimization Problems Using Large Language Models. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

#### HONORS AND AWARDS

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|-------------------------------------------------------------------------------|------|
| People's Choice Award, Global AI Solutions Expo, Amazon                       | 2026 |
| INFOR Best Paper Award, 67th CORS Annual Conference                           | 2026 |
| Bilsland Dissertation Fellowship, Purdue University                           | 2026 |
| Purdue Graduate Student Government (PGSG) Travel Grant                        | 2025 |
| Ryland Travel Grant, School of Chemical Engineering, Purdue University        | 2025 |
| Purdue Graduate Student Government (PGSG) Travel Grant                        | 2024 |
| Chao Travel Grant, School of Chemical Engineering, Purdue University          | 2024 |
| Nottingham Engineering Excellence Scheme (Top 1.0%), University of Nottingham | 2019 |
| British Petroleum Prize, University of Nottingham                             | 2019 |
| Provost's Scholarship (Top 1.5%), University of Nottingham                    | 2018 |
| Dean's Scholarship (Top 10%), University of Nottingham                        | 2017 |

#### PROFESSIONAL ACTIVITIES

Session Chair: *INFORMS Annual Meeting* (2026)

Reviewer: *NeurIPS* (2026); *Manufacturing & Service Operations Management* (2026); *Computers & Chemical Engineering* (2025-2026); *Digital Chemical Engineering* (2025-2026); *Applied Energy* (2025)

#### SKILLS

Languages: Python, Julia, SQL

ML/LLM Stack: PyTorch, PyG, SB3, HuggingFace, Google ADK, Claude Agent SDK

OR Stack: JuMP, Gurobi, Xpress, Pyomo, CPLEX

Others: AWS EC2, ECS, Docker, Lambda, DynamoDB, S3